

Deriving Black Hole Entropy

from the homogeneity of units

Year 11 investigation • Physics & Mathematics • 31 marks

What you are aiming for. Stephen Hawking and Jacob Bekenstein found that a black hole carries entropy

$$S = \frac{\pi A k_B c^3}{2 h G}.$$

In this investigation you will recover almost the whole of that formula without doing any of the hard physics, using only the requirement that both sides of an equation must share the same units. You will build the units of every ingredient from its defining equation, fix the power of each quantity, and see precisely which part of the formula this method can never reach. Everything you need is contained in these pages, so work through them in order.

Preamble: the SI base units

Every mechanical and thermal quantity in this task is built from four base units, which we track as abstract *dimensions* so that the algebra stays clean.

Quantity	Base unit	Symbol	Dimension
length	metre	m	L
mass	kilogram	kg	M
time	second	s	T
temperature	kelvin	K	Θ

Reading a dimension. A speed is a length divided by a time, so its dimension is $L T^{-1}$ and its unit is m s^{-1} . The square brackets $[X]$ are read “the dimension of X ”. Two quantities can be added, subtracted, or set equal *only* when their dimensions agree; this single rule, called *dimensional homogeneity*, is the whole engine of the investigation.

Section A | What entropy actually counts

Entropy as a count of arrangements

A pack of 52 playing cards can be ordered in

$$52! = 52 \times 51 \times \cdots \times 2 \times 1 \approx 8 \times 10^{67}$$

different ways. If all you are told is “it is this pack of cards”, then as far as you are concerned the pack could be in any one of those $52!$ arrangements; the number of arrangements consistent with what you know is written Ω . Ludwig Boltzmann proposed that the entropy of such a system is

$$S = k_B \ln \Omega,$$

so a freshly shuffled pack, with $\Omega = 52!$, carries a large entropy, whereas a pack you have just sorted into a known order has $\Omega = 1$ and therefore $S = k_B \ln 1 = 0$. Entropy, on this reading, measures how much you do *not* know about the exact internal arrangement once the overall description has been fixed.

How large is Ω , and how large is its logarithm? Multiplying $52!$ out by hand is hopeless, and in any case it is the logarithm $\ln \Omega$ on which the entropy actually depends, so we lean on Stirling’s approximation, which states that for large n

$$n! \approx \sqrt{2\pi n} \left(\frac{n}{e}\right)^n, \quad \text{equivalently} \quad \ln n! \approx n \ln n - n + \frac{1}{2} \ln(2\pi n).$$

Setting $n = 52$, the leading part $52 \ln 52 - 52 \approx 153.5$ together with the small correction $\frac{1}{2} \ln(2\pi \cdot 52) \approx 2.9$ gives $\ln 52! \approx 156.4$, which both recovers the headline figure $\Omega = 52! = e^{\ln 52!} \approx e^{156.4} \approx 8 \times 10^{67}$ and fixes the entropy of a shuffled pack at $S = k_B \ln \Omega \approx 156 k_B$. Read as information this is roughly $156 / \ln 2 \approx 226$ bits, so a little over two hundred yes/no questions would pin down the exact order; and it is this same device, with the leading term $\ln n! \approx n \ln n - n$ doing the work, that lets one handle the vastly larger counts of microstates arising for a gas or for a black hole, where listing the arrangements is wholly out of the question.

The same idea in bits. Claude Shannon described the same uncertainty as information. If Ω arrangements are equally likely, the number of yes/no questions needed to pin down which one you have is

$$H = \log_2 \Omega \quad \text{bits},$$

and the two pictures are joined by $S = k_B \ln \Omega = (k_B \ln 2) H$, so thermodynamic entropy is simply Shannon’s information measured in units of $k_B \ln 2$ for every bit. Notice that the count Ω and the bit count H are pure numbers carrying no units at all; it is the constant k_B that supplies the units of J K^{-1} , and you should keep that fact in mind, because it returns with some force in Section D.

1. (3 points) A black hole of horizon area A is found to have entropy S . Using the ideas above, answer briefly.
 - (a) (1 point) A second black hole has a larger horizon area. State, with a one-line reason, whether you expect it to have more or less entropy.

- (b) (1 point) Explain why the count Ω carries no units, while S does.

- (c) (1 point) Write down the dimension $[S]$ of entropy, given that S is measured in J K^{-1} and that $\text{J} = \text{kg m}^2 \text{s}^{-2}$.

How large is that entropy? Stirling's approximation

The number $\Omega = 52!$ is far too large to write out in full, yet the entropy depends on it only through its logarithm $\ln \Omega$, and there is a celebrated formula, due to James Stirling, that gives the factorial of a large number without ever multiplying its terms together:

$$n! \approx \sqrt{2\pi n} \left(\frac{n}{e}\right)^n, \quad \text{equivalently} \quad \ln n! \approx n \ln n - n + \frac{1}{2} \ln(2\pi n).$$

The dominant part is $n \ln n - n$, with the final term sharpening it. Applied to a pack of cards this turns the unmanageable $52!$ into a modest number, and it is worth noticing how completely the logarithm tames the size, since a quantity close to 10^{68} becomes, after its logarithm is taken, a number you could hold in your hand. This is the very reason entropy is built on $\ln \Omega$ rather than on Ω itself, for the logarithm also turns the multiplying-together of independent systems into plain addition, so that their entropies simply add.

2. (3 points) Use Stirling's approximation to put a number to the entropy of a freshly shuffled pack.

- (a) (2 points) Show that $\ln(52!) \approx 156$, taking $\ln 52 \approx 3.95$ and $\ln(2\pi \times 52) \approx 5.79$.

- (b) (1 point) Hence give the entropy of the shuffled pack as a multiple of k_B , and, dividing by $\ln 2 \approx 0.693$, the number of yes/no questions (bits) needed to identify the exact ordering.

Section B | The toolkit of constants

Each constant of nature below arrives through a defining equation. Most of those equations are statements of *proportionality*, in which one quantity grows in step with another and the constant of nature is simply the number that turns the words “is proportional to” into an equals sign; one of them, the law of gravitation, also carries a power of distance that comes straight from the geometry of how an influence spreads through space. Before the units of the constants can be read, the dimensions of a few ordinary quantities that appear inside those equations are needed, so those are built first and supplied here for you to use.

Foundations: dimensions you will build on

Each of these follows from a definition you already know, and each will be needed below.

- **Acceleration** is the rate of change of velocity, a velocity divided by a time, so $[a] = [v] / [t] = (\text{L T}^{-1}) / \text{T} = \text{L T}^{-2}$ (m s^{-2}).
- **Force** obeys Newton’s second law $F = ma$, force being proportional to acceleration with the mass as the constant of proportionality, so $[F] = [m] [a] = \text{M L T}^{-2}$ (the newton).
- **Energy** can be read from kinetic energy $E_{\text{kin}} = \frac{1}{2}mv^2$, where m is the mass and v the speed, so $[E] = [m] [v]^2 = \text{M} (\text{L T}^{-1})^2 = \text{M L}^2 \text{T}^{-2}$ (the joule). The $\frac{1}{2}$ is a pure number and contributes nothing to the dimension.
- **Frequency** is a count of oscillations per unit time; a count is a pure number, so $[f] = \text{T}^{-1}$ (the hertz).

3. (2 points) **The speed of light, c .** When an object moves at a steady speed the distance it travels grows in direct proportion to the time for which it has been moving, so that travelling for twice as long carries it twice as far; this proportionality $d \propto t$ becomes an equation once a constant of proportionality is inserted, and that constant is precisely the speed. Light in a vacuum is found always to move at the same fixed speed whatever its source or direction, so this particular constant of proportionality is one of the fundamental constants of nature.

$$d = ct, \quad \text{equivalently} \quad c = \frac{d}{t},$$

where d is the distance travelled (in m) and t is the time taken (in s). Find the dimension $[c]$.

4. (2 points) **The Boltzmann constant, k_B .** The temperature of a gas measures how vigorously its molecules move, and experiment shows that the average translational kinetic energy of a molecule rises in direct proportion to the absolute temperature, so that doubling the temperature doubles that average energy. Writing this proportionality $\overline{E}_{\text{kin}} \propto T$ as an equation introduces a constant of proportionality, the Boltzmann constant, together with a pure-number factor of $\frac{3}{2}$ that arises because a molecule may move independently along three perpendicular directions and contributes $\frac{1}{2}k_B T$ to each.

$$\overline{E}_{\text{kin}} = \frac{3}{2} k_B T,$$

where $\overline{E}_{\text{kin}}$ is the average translational kinetic energy of one molecule (in J) and T is the absolute temperature (in K); the factor $\frac{3}{2}$ is a pure number. Rearrange for k_B and find its dimension $[k_B]$.

5. (2 points) **The Planck constant, h .** The energy carried by a single particle of light depends on the colour of that light, with bluer, higher-frequency light delivering more energetic photons than redder light; the relationship is again one of direct proportion, $E \propto f$, and the constant of proportionality that converts a frequency into an energy is Planck's constant.

$$E = h f,$$

where E is the energy of one photon (in J) and f is the frequency of the light, the number of oscillations each second (in Hz, equivalently s^{-1}). Rearrange for h and find its dimension $[h]$.

6. (2 points) **The gravitational constant, G .** Two masses attract one another, and the strength of that attraction is proportional to each of the masses separately, so that doubling either mass doubles the force and the two together enter as the product $m_1 m_2$. The way the force weakens with distance comes instead from geometry: imagine the gravitational influence of a mass streaming outward equally in every direction, so that by the time it reaches a distance r it is spread evenly over the surface of a sphere of radius r . That surface has area $4\pi r^2$, and since the same total influence is now shared across an area growing as the square of the distance, its strength at any one point must fall as $1/r^2$; this is the inverse-square law, the very same spherical-spreading argument that governs how the brightness of a lamp or the intensity of any radiation diminishes with distance. Combining the two ideas gives $F \propto m_1 m_2 / r^2$, and the constant of proportionality, with the geometric factor absorbed into it, is Newton's gravitational constant.

$$F = \frac{G m_1 m_2}{r^2},$$

where F is the gravitational force (in N), m_1 and m_2 are the two masses (in kg), and r is the distance between their centres (in m). Rearrange for G and find its dimension $[G]$.

7. (2 points) **The horizon area, A .** Area is a two-dimensional quantity, and for any family of shapes of the same form it scales as the square of a chosen length, so that doubling every length quadruples the area; this is a power law, $A \propto \ell^2$. The horizon of a black hole is a sphere, and so

its area follows the same spherical geometry met above for gravity, $A = 4\pi r^2$, with the 4π once more a pure number.

$$A = \ell^2,$$

where ℓ is a characteristic length of the shape (in m); for the spherical horizon $\ell = r$, its radius. Find the dimension $[A]$.

Collect your results. Transcribe the five dimensions into this table; you will draw on it throughout Section D.

Quantity	Symbol	Dimension
entropy	S	
speed of light	c	
Boltzmann constant	k_B	
Planck constant	h	
gravitational constant	G	
horizon area	A	

Section C | The method, in the abstract

Recovering a law from its units

Suppose a quantity Q is built as a product of powers of some others, $Q = \kappa X_1^{p_1} X_2^{p_2} \dots$, where κ is a pure number. Because the two sides must be dimensionally homogeneous, the exponent of each base dimension (M, L, T, Θ) on the right must equal its exponent on the left. That gives one equation per base dimension, and solving the resulting simultaneous equations fixes the powers $p_1, p_2, \dots$. The pure number κ , however, has no dimension of its own, so this argument can never reveal it.

8. (2 points) Before applying the method, answer in one sentence each.
- (a) (1 point) We have four base dimensions and will introduce five unknown powers in Section D. Explain why four equations alone cannot determine five unknowns, and what kind of extra information would be needed.

- (b) (1 point) State why a dimensionless prefactor such as $\pi/2$ is invisible to this method.

Section D | Building and solving the equation

We assume the entropy is a product of powers of the constants in your table together with a pure number κ :

$$S = \kappa k_B^a A^b c^d h^e G^f.$$

9. (3 points) **The temperature argument.** Look only at the temperature dimension Θ .

- (a) (2 points) Among k_B, A, c, h, G , identify which carry the dimension Θ , and hence write down the equation that the temperature dimension imposes.

- (b) (1 point) Solve for a , and explain why this is the result anticipated at the end of Section A.

10. (6 points) **The remaining three dimensions.** With $a = 1$ fixed, match the dimensions M, T and L in turn.

- (a) (3 points) Write down the three equations obtained by matching M, T and L, and simplify each using $a = 1$.

- (b) (1 point) Show that these reduce to $e = f$, $d = -3e$ and $b = -e$, a family of solutions controlled by the single power e .

- (c) (2 points) **Bekenstein's physical input.** Bekenstein argued that the entropy is proportional to the horizon *area*, i.e. to the first power of A . Use $b = 1$ to fix e , and hence write down all of a, b, d, e, f .

11. (2 points) Substitute your powers back into the assumed form to obtain the entropy formula up to the unknown number κ .

Section E | The number you cannot reach

12. (2 points) The full quantum calculation gives $\kappa = \pi/2 \approx 1.571$.
- (a) (1 point) Explain why your method, however carefully carried out, could never have produced the value $\pi/2$.

- (b) (1 point) Your result shows $S \propto A$ (surface area) rather than $S \propto V$ (volume). Name the broad principle that this surface-scaling of information supports, in a sentence.